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Rastereasy: A Python package for an easy manipulation of remote sensing images

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Software

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Summary

The analysis and processing of remote sensing images have many important applications in various fields such as environmental monitoring, urban planning, or even agriculture. However, handling large georeferenced raster datasets can be challenging due to their complexity and size.

rastereasy is a Python library for simple manipulation of georeferenced images (`*.tif`, `*.jp2`, `*.shp`, …) [Ritter & Ruth (1997)](Mamatov et al., 2024). The goal is to simplify geospatial workflows by offering tools for reading and processing raster and vector files, resampling, cropping, reprojecting, stacking, etc of raster images, easy visualizations such as color composites and spectral plots, use (train / apply) some classical Machine Learning algorithms on images …

Compared to traditional RGB image manipulation, satellite images are highly specific due to their specific notions of spatial resolution, geographic extent, projection system, and they embed multiple spectral bands which prevents from an easy vizualisation. Dedicated software such as QGIS exists to handle these images, as well as specialized libraries for tasks like tiling, resampling, and reprojection. However, these tools require expertise in metadata management and geospatial systems, which can be a barrier for users unfamiliar with geographic data handling.

rastereasy is designed to simplify these processes, providing an easy-to-use interface for standard operations on multispectral and georeferenced images. It is particularly aimed at users who are experienced in data processing but not necessarily in geospatial analysis, while also streamlining workflows for geographers by leveraging `rasterio` and other geospatial libraries. It is particularly useful, among other things, for preparing sample data for deep neural networks.

The source code is available at https://github.com/pythonraster/rastereasy and a documentation https://rastereasy.github.io/.

Statement of need

Many existing remote sensing libraries, such as `rasterio` and `gdal` [Garrard (2016)](Gillies et al., 2013), provide powerful functionalities but often require a deep understanding of geospatial data structures. `rastereasy` abstracts these complexities by offering a high-level interface for:

- **Band manipulation**: Extract, reorder, and remove spectral bands easily.
- **Tiling and stitching**: Split large raster images into smaller tiles and reconstruct them.
- **Harmonization**: Align rasters with different spatial resolutions and extents.

- **Visualization tools**: Quick and interactive display of georeferenced images and spectral signatures.
- **Basics of machine learning**: Clustering of images (Ikotun et al., 2023), adaptation of spectral bands (domain adaptation) (Courty et al., 2016)
- **Fusion of classifications**: Fusion of mass function under the Dempster-Shafer framework (Shafer, 1992)
- …

The package is designed for researchers and practitioners in remote sensing who need efficient tools for image preprocessing and analysis. It integrates seamlessly with `rasterio` and `numpy`, making it compatible with existing geospatial workflows.

Example of use

In rastereasy, the core class of the library is **Geoimage**. This class allows users to manipulate a satellite image as a `numpy` array while preserving essential geospatial information, such as georeferencing, spectral bands, and projection system. This makes it easy to perform calculations on the data while maintaining its spatial consistency.

For example, applying a simple transformation, extracting spectral bands, performing operations or modifying an image is straightforward: Here's a quick example of what you can do with rastereasy:

```python
import rastereasy

# Load a georeferenced image
image = rastereasy.Geoimage("example.tif")

# Get image information
image.info()

# Print value of pixel [100,200]
print(image[100,200])

# Create a color composite
image.colorcomp(['4', '3', '2'])

# Resample and reproject
image_resampled = image.resampling(2)
image_reproject = image.reproject("EPSG:4326")

# This can also be done in inplace mode
image.resampling(2, inplace=True)

# Save the processed image
image.save("processed_image.tif")
```

all these functions have an `inplace` option to modify directly the images

Band Operations and feature computation

Users can easily manipulate spectral bands using high-level functions and compute indices (Xue & Su, 2017):

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```python
import rastereasy

# Load a georeferenced image
img = rastereasy.Geoimage("example.tif")

# select red and near-infrared bands, positined in 4th and 8th positions
r=img.select_bands(4)
nir=img.select_bands(8)

# Compute NDVI (Normalized Difference Vegetation Index )
NDVI = (nir-r)/(nir+r)

# Apply a simple transformation: remove specific spectral bands
img = img.remove_bands([10, 8])

# Perform a reprojection
img_reproj = img.reproject(target_crs="EPSG:4326")
```

In one prefers to deal with explicit names for spectral bands, this is easily done by specifying names

```python
import rastereasy

# Load a satellite image and give specific names
name_bands = {"NIR":8,"G":3,"CO" : 1,"SWIR2":11,"B": 2,
              "R":4,"RE1":5,"RE2":6,"RE3":7,"WA":9,
              "SWIR1":10,"SWIR3":12}
img = Geoimage("satellite_image.tif",names=name_bands)

# select red and near-infrared bands
r=img.select_bands('R')
nir=img.select_bands('NIR')

# Compute NDVI (Normalized Difference Vegetation Index )
NDVI = (nir-r)/(nir+r)

# Apply a simple transformation: remove specific spectral bands
img_removed = img.remove_bands(["SWIR1", "NIR"])

# Perform a reprojection
img_reproj = img.reproject(target_crs="EPSG:4326")

# see also get_bands, switch_bands, ...
```

Image Tiling

Splitting a large image into smaller tiles with optional overlap (useful for data preparation):

```python
from rastereasy import im2tiles
im2tiles("satellite_image.tif", "output_folder", nb_lig=512, nb_col=512, overlap=50)
```

Visualization

One can visualize histogrmas, color composites, spectra, …

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```
import rastereasy
image = rastereasy.Geoimage("example.tif")
# plotting spectra
image.plot_spectra()
# Making a color composition
image.colorcomp([4,3,2])
# Visualization of histograms
image.hist(superpose=True)
```

This gives the following images:

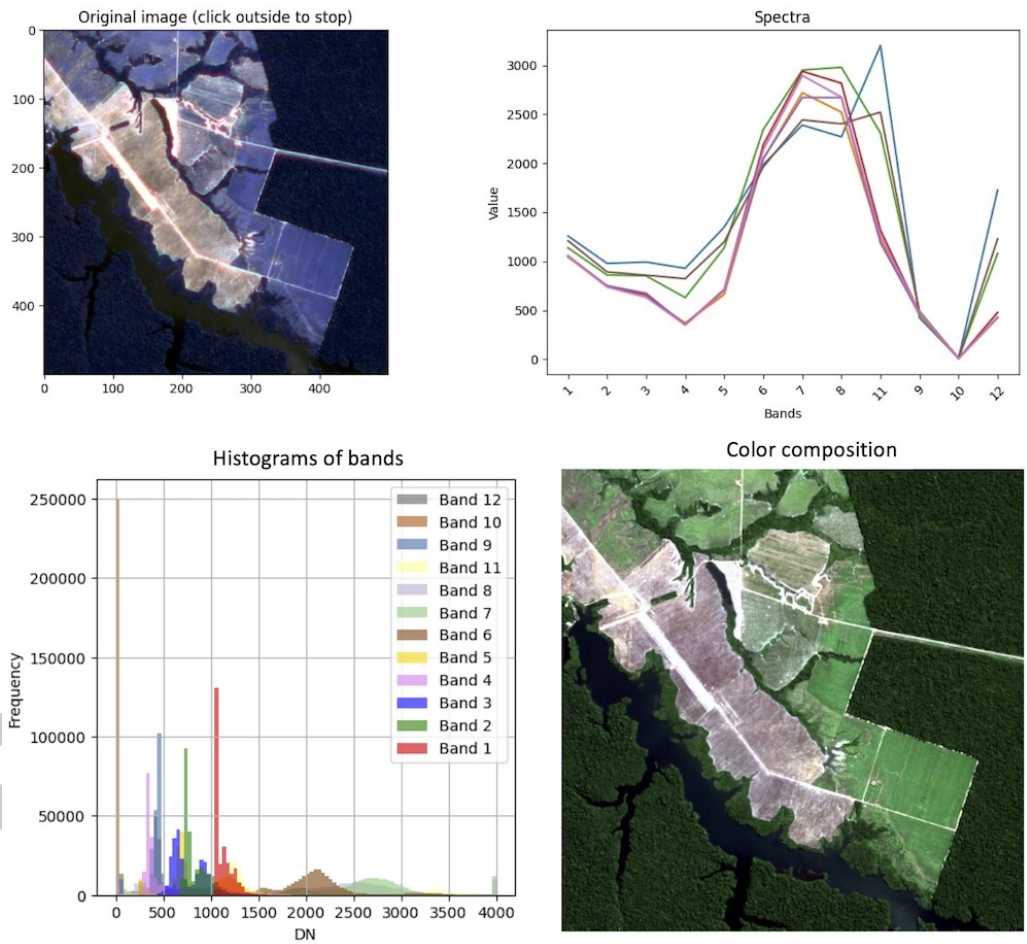

Figure 1: Examples of visualizations provided by rastereasy. Complete examples can be seen on the rastereasy package documentation : https://rastereasy.github.io/

Harmonization

Aligning images with different extents and resolutions:

```
from rastereasy import extract_common_areas
im1_common,im2_common = extract_common_areas(im1, im2)
```

Adapt the spectral values of to images with optimal transport

```
import rastereasy
```

```
image1 = rastereasy.Geoimage("im1.tif")
image2 = rastereasy.Geoimage("im2.tif")
# Change image 1 to adapt it to image 2

image1.adapt(image2, mapping='sinkhorn', inplace=True)
```

Performance and Scalability

`rastereasy` leverages numpy for efficient numerical operations and `rasterio` for optimized I/O operations, ensuring scalability for large datasets. Parallel processing capabilities are planned for future releases.

A complete documentation is available at https://rastereasy.github.io/ with many notebooks for examples.

The source code is available here : https://github.com/pythonraster/rastereasy

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